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AWS Services: A Simple & Broad Guide

Introduction

AWS (Amazon Web Services) is like a giant digital store. Instead of buying physical computers, hard drives, and cables, you rent them from Amazon over the internet. You only pay for what you use, just like your water or electricity bill at home.

1. Compute (The Brains)

Amazon EC2

What it is: A virtual computer.

In simple words: Instead of buying a laptop or a server box, you rent a computer from Amazon. You can install anything you want on it. If you need a more powerful one, you can upgrade it with one click.

AWS Lambda

What it is: A computer that only works when needed.

In simple words: You don't rent the whole computer. You just give Amazon your code. The code only runs when someone uses your app. If no one is using it, you pay \$0.

2. Storage (The Digital Closet)

Amazon S3

What it is: A giant storage bucket for files.

In simple words: This is where you save photos, videos, and documents. It never runs out of space. It is like an infinite Google Drive for developers.

Amazon EBS

What it is: A virtual hard drive.

In simple words: This is a hard drive that you plug into your EC2 computer. It is used for things that need to be fast, like running an operating system.

3. Databases (The Organizers)

Amazon RDS

What it is: A managed filing system for data.

In simple words: It keeps your data (like customer names and passwords) in organized tables. Amazon does the hard work of backing up the data so you don't lose it.

Amazon DynamoDB

What it is: A super-fast digital notebook.

In simple words: It is a database designed for speed. Even if millions of people use your app at the same time, it finds information instantly.

4. Networking (The Roads)

Amazon VPC

What it is: Your own private network.

In simple words: It is like having a private, fenced-in backyard in a big public park. It keeps your computers safe from people you don't want in your network.

Amazon Route 53

What it is: The internet's phone book.

In simple words: It connects website names (like `www.example.com`) to the actual computer address. It helps users find your website easily.

5. Delivery and Security

Amazon CloudFront

What it is: A fast delivery service.

In simple words: It sends your website's photos and videos to servers near the user. If a user is in London, they get the file from London, not from a far-away country.

AWS IAM

What it is: The security guard.

In simple words: This is where you decide who can log in. You can give 'keys' to employees so they can only see the parts of the account they need for their job.

Compute Services

Service	What it does	How it works
EC2	Rents you virtual computers (servers) in cloud	You pick the size(small,/medium/large), choose Windows or Linux, and pay only for the hours you use.
Lambda	Runs your code without managing any servers	You upload your code, and AWS runs it automatically when something triggers it(like a file upload or website visit)
ECS	Runs containers (lightweight app packages)	You tell AWS where to run your app, it handles the rest.

EKS	Runs Kubernetes (a tool that manages containers)	AWS manages the complex Kubernetes setup so you don't have to.
Fargate	Runs containers without managing the underlying servers	You just say "run this container" and AWS figures out where.
Elastic Beanstalk	Deploys your website/app automatically	You upload your code, AWS automatically sets up servers, load balancers, and scaling.
Batch	Runs thousands of calculations at once	You submit a list of tasks, AWS runs them when computers are free.
Outposts	Brings AWS services to your own physical office	AWS installs a small server rack in your building that behaves like the cloud.

Storage Services

Service	What it does	How it works
RDS	Runs normal databases (MySQL, PostgreSQL, SQL Server)	AWS sets up the database, does automatic backups, and handles updates for you.
Aurora	A faster, better version of MySQL/PostgreSQL	5x faster than normal databases, but works the same way.
DynamoDB	Super-fast NoSQL database for any scale	Works like giant spreadsheet, you give it a key, it gives you data instantly
ElasticCache	Stores temporary data super fast (in memory)	Like a sticky note on your desk, very fast but data disappears if power goes off.
Redshift	Analyzes billions of rows of data quickly	Spreads data across many computers so queries run in seconds instead of hours.
Neptune	Stores connected data (like social networks)	Works well for "friend connections" or "road maps", follows relationships easily.
DocumentDB	Stores JSON documents (like MongoDB)	Each record can have different fields, flexible like a filing cabinet.

Networking Services

Service	What it does	How it works
VPC	Creates you own private network inside AWS	Like having your own room in a big building, you decide who comes in
CloudFront	Makes your website load fast everywhere in the world	Copies your files to 400+ locations worldwide so users download from nearest one.
Route 53	Translates website names to IP addresse	Like a phonebook for the internet, routes visitors to your servers
API Gateway	Creates entry points for your office to AWS	Like a reception desk that accepts requests and sends them to the right place.
Direct Connect	Private fiber cable from your office to AWS	No internet, direct, fast, secure connection
ELB	Spreads traffic across many servers	If one server gets busy, sends new visitors to less busy servers.
Global Accelerator	Makes traffic from users to you faster	Finds the fastest path across the internet automatically.